

Q1. Create a tuple:

```
my_tuple = (10, 20, 30, 40, 50)
```

Print:

- First element
- Last element
- Length of the tuple
- Elements from index 1 to 3

ANS

```
# Create a tuple
```

```
my_tuple = (10, 20, 30, 40, 50)
```

```
# First element
```

```
print("First element:", my_tuple[0])
```

```
# Last element
```

```
print("Last element:", my_tuple[-1])
```

```
# Length of the tuple
```

```
print("Length of the tuple:", len(my_tuple))
```

```
# Elements from index 1 to 3
```

```
print("Elements from index 1 to 3:", my_tuple[1:4]).
```

2. Create a tuple: fruits = ("apple", "banana", "mango", "orange") Print: ● Second fruit ● Last two fruits ● Total number of fruits

ANS

```
# Create a tuple
```

```
fruits = ("apple", "banana", "mango", "orange")
```

```
# Second fruit
```

```
print("Second fruit:", fruits[1])
```

```
# Last two fruits
```

```
print("Last two fruits:", fruits[-2:])
```

```
# Total number of fruits
```

```
print("Total number of fruits:", len(fruits))
```

3. Create a set: numbers = {10, 20, 30, 40, 50} Print: • Complete set • Length of the set • Check whether 30 is present in the set

ANS

```
# Create a set
```

```
numbers = {10, 20, 30, 40, 50}
```

```
# Print complete set
```

```
print("Complete set:", numbers)
```

```
# Print length of the set
```

```
print("Length of set:", len(numbers))
```

```
# Check whether 30 is present in the set
```

```
if 30 in numbers:
```

```
    print("30 is present in the set")
```

```
else:
```

```
print("30 is not present in the set")
```

4. Create two sets: set1 = {1, 2, 3, 4} set2 = {3, 4, 5, 6} Find: • Union • Intersection

ANS

```
# Create two sets
```

```
set1 = {1, 2, 3, 4}
```

```
set2 = {3, 4, 5, 6}
```

```
# Union
```

```
print("Union:", set1.union(set2))
```

```
# Intersection
```

```
print("Intersection:", set1.intersection(set2))
```

• Intersection 5. Create a dictionary: student = {"name": "Kriti", "age": 20, "course": "Python"} Print: • Complete dictionary • Student name • Student age • Student course Create a list: numbers = [12, 45, 7, 23, 56, 89, 34] Write a program to find: o Largest element o Second largest element o Smallest element

ANS

```
# Create a dictionary
```

```
student = {
```

```
    "name": "Kriti",
```

```
    "age": 20,
```

```
    "course": "Python"
```

```
}
```

```
# Print details
```

```
print("Complete Dictionary:", student)
```

```
print("Student Name:", student["name"])
```

```
print("Student Age:", student["age"])
```

```
print("Student Course:", student["course"])
```

6. Largest, Second Largest, and Smallest Element

```
# Create a list
```

```
numbers = [12, 45, 7, 23, 56, 89, 34]
```

```
# Sort the list
```

```
numbers.sort()
```

```
# Find elements
```

```
smallest = numbers[0]
```

```
second_largest = numbers[-2]
```

```
largest = numbers[-1]
```

```
print("Largest Element:", largest)
```

```
print("Second Largest Element:", second_largest)
```

```
print("Smallest Element:", smallest)
```

6. Create a list: arr = [10, 20, 30, 40, 50, 60] Write a function that takes the list as input and returns the list in reverse order without using the reverse() method.

ANS

```
# Function to reverse a list without using reverse()
```

```

def reverse_list(arr):

    return arr[::-1]


# Create a list

arr = [10, 20, 30, 40, 50, 60]


# Call function

reversed_arr = reverse_list(arr)


print("Original List:", arr)

print("Reversed List:", reversed_arr)

```

Without slicing ([::-1]):

```

def reverse_list(arr):

    rev = []

    for i in range(len(arr)-1, -1, -1):

        rev.append(arr[i])

    return rev


arr = [10, 20, 30, 40, 50, 60]

print(reverse_list(arr))

```

7. Create a tuple: data = (5, 10, 15, 20, 25, 30, 35) Write a program to: ○ Count how many elements are divisible by 5 ○ Find the sum of all elements ○ Find the average of the tuple

ANS

```

# Create a tuple

data = (5, 10, 15, 20, 25, 30, 35)

```

```
# Count elements divisible by 5
```

```
count = 0
```

```
for num in data:
```

```
    if num % 5 == 0:
```

```
        count += 1
```

```
# Find sum
```

```
total = sum(data)
```

```
# Find average
```

```
average = total / len(data)
```

```
# Print results
```

```
print("Count divisible by 5:", count)
```

```
print("Sum of all elements:", total)
```

```
print("Average:", average)
```

8. Create a dictionary: students = { "Aman": 78, "Riya": 92, "Kriti": 88, "Rahul": 95 } Write a program to:

- Find the student with the highest marks
- Find the student with the lowest marks
- Print only the students who scored more than 85 marks

ANS

```
# Create a dictionary
```

```
students = {
```

```
    "Aman": 78,
```

```
    "Riya": 92,
```

```

    "Kriti": 88,
    "Rahul": 95
}

# Student with highest marks
highest_student = max(students, key=students.get)

# Student with lowest marks
lowest_student = min(students, key=students.get)

print("Highest Marks:")
print(highest_student, "-", students[highest_student])

print("\nLowest Marks:")
print(lowest_student, "-", students[lowest_student])

print("\nStudents scoring more than 85 marks:")
for name, marks in students.items():
    if marks > 85:
        print(name, "-", marks)

```

9. Write a function: `count_frequency(arr)` that takes a list as input and prints the frequency of each element. Example: `arr = [1, 2, 2, 3, 1, 4, 2]` Output: 1 -> 2 times 2 -> 3 times 3 -> 1 time 4 -> 1 time

ANS

```

# Function to count frequency of elements

```

```

def count_frequency(arr):
    freq = {}

    for item in arr:
        if item in freq:
            freq[item] += 1
        else:
            freq[item] = 1

    for key, value in freq.items():
        print(key, "->", value, "times")

# Example list
arr = [1, 2, 2, 3, 1, 4, 2]

count_frequency(arr)

```

10. Write a function: `find_duplicates(arr)` that takes a list as input and prints all duplicate elements present in the list. Example: `arr = [10, 20, 30, 20, 40, 10, 50, 30]` Output:
Duplicate elements are: 10 20 30

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```

# Function to find duplicate elements

```

```

def find_duplicates(arr):
    duplicates = set()

```

```
for item in arr:
```

```
    if arr.count(item) > 1:
```

```
        duplicates.add(item)
```

```
print("Duplicate elements are:")
```

```
for item in duplicates:
```

```
    print(item)
```

```
# Example list
```

```
arr = [10, 20, 30, 20, 40, 10, 50, 30]
```

```
find_duplicates(arr)
```


